

ABDULRAHMAN OMAR

AI Engineer | LLM Systems | RAG Architectures | Multi-Agent Systems

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Summary

AI Engineer and startup CTO with hands-on experience building production-grade LLM applications, RAG systems, and multi-agent architectures. Experienced across the full lifecycle: fine-tuning and self-hosting open-source LLMs (QLoRA, vLLM, Modal), building hybrid-retrieval RAG pipelines, and orchestrating autonomous agents behind FastAPI services, from research prototyping to production deployment.

Publications

Abdulrahman Omar, Mayada Hadhoud (2026). “Explaining the Unexplainable? A Ground-Truth Benchmark of Classical, Concept-Based, and Mechanistic XAI on Financial Forecasters.” *Preprint, under review* (first author). | [GH](#) Code & Paper

- Designed a synthetic ground-truth faithfulness benchmark that tests whether XAI attributions recover known feature importance; evaluated 23 explainability methods (SHAP, LIME, ALE, Integrated Gradients, TCAV, sparse autoencoders, circuit analysis, GNNExplainer, and more) across 16 ML/DL models on a leakage-free, near-zero-signal financial forecaster.
- Applied stock-clustered bootstrap inference with explicit effective-sample-size and minimum-detectable-effect reporting; found that popular attribution methods diverge sharply under feature collinearity. Released a fully reproducible pipeline on Modal, using distributed **Apache Spark** to ingest a 23 GB financial-news corpus.

Professional Experience

AI Engineer

BrightBrainsIT

Aug 2026 – Present

Dubai, UAE (Remote)

- Building an AI-powered Digital Employee Experience (DEX) platform on Nextthink (Infinity & V6 APIs) for a UAE government client (eia.gov.ae), spanning four pillars: conversational AI assistant, playbook-driven automation engine, unified insights dashboard, and proactive anomaly detection.
- Built a natural-language analytics assistant that compiles user questions into guarded NXQL queries (schema-linked planning → AST → lint → execution), achieving 77% execution match on an internal benchmark, with measured refusals instead of guesses for unanswerable queries.
- Hardened LLM reliability on a 7B open-source model via structured JSON outputs with retries, self-consistency voting, deterministic schema matching in code, and replaying past query-engine rejections as learned “case law”.
- Implemented multi-step “why” investigations that run playbooks of correlated queries, plus a documentation-grounded lane answering how-to questions from vendored product docs with quoted, cited sentences.
- Designed a human-in-the-loop remote-action flow: the assistant proposes fixes from an admin-configured action catalog, and every write passes one audited gate with dry-run defaults, blast-radius caps, and post-run verification.

Co-Founder & CTO

Everyside – Arabic-Native AI Marketing OS

Jul 2025 – Present

Remote (Egypt & Saudi Arabia)

- Co-founded an Arabic-native AI Marketing OS for business owners, media buyers, and agencies in Egypt and Saudi Arabia: define your brand once, and the platform designs, writes, and launches on-brand campaigns in Arabic and English; nothing publishes without human approval.
- Architected and built the full production platform: FastAPI + Pydantic v2 backend with LLM-driven on-brand generation (pydantic-ai over Claude/OpenAI), Redis-backed background jobs for rendering and publishing, Supabase (Postgres/Auth/Storage), and Paymob billing, deployed on GCP Cloud Run.
- Built ad-channel publishing integrations (Meta, TikTok, Snap) and a bilingual Next.js / React product UI with RTL-aware EN/AR i18n (Arabic authored), enforced by CI checks.
- Re-architecting the platform for multi-tenancy so agencies can run multiple client brands under one account, while driving the close of a funding round with a regional investor.

AI Engineer (Part-time)

Snippet – Recruited by Mostafa Touny

Jan 2026 – Mar 2026

Remote

- Architected hybrid AI system combining deterministic guardrails with LLM-based generation for knowledge graph consistency, enabling semantic search across 20M+ documents.
- Designed and implemented conflict-resolution pipelines for multi-source knowledge ingestion using Neo4j, handling complex entity resolution and citation verification workflows.
- Built graph-based semantic search with context-aware retrieval using RAG architectures and semantic embeddings for large-scale technical documentation.
- Led end-to-end technical strategy from requirements translation to production deployment; validated on documentation from the EndeavourOS Linux project.

AI Engineer Intern

Aug 2025 – Oct 2025

Optomatica – Recruited by Dr. Mohammed El-Beltagy

Cairo, Egypt

- Built advanced RAG architecture for ActiveQ.ai (an enterprise RFP automation platform) with query expansion, semantic chunking, and hybrid retrieval (dense + sparse) with LLM reranking, improving accuracy by +15%.
- Developed RAG evaluation framework using Ragas for automated benchmarking of retrieval quality and factual consistency.

Data Science Intern (Part-time)

Aug 2025 – Oct 2025

Sigma EMEA

Remote (1 day/wk on-site)

- Built customer segmentation and recommendation engine for telecom operator processing 2M+ customer records.
- Led team of 5 delivering predictive analytics solution.

Technical Projects

Everyside: Arabic-Native AI Marketing OS | *FastAPI, Next.js, Supabase, GCP*

2025 – Present

- Co-founded and built end to end an AI platform that designs, writes, and launches on-brand campaigns in Arabic and English, with human approval before anything publishes; full detail under Professional Experience.

Pulse: AI Multi-Agent Marketing Platform | *CrewAI, DSPy, PyTorch, Supabase, Redis, Celery, vLLM, Docker* 2025

- **Creator & Lead Engineer.** Led a team of 4 and architected a production-grade multi-agent AI platform for social-media marketing using CrewAI and DSPy, orchestrating specialized autonomous agents (content creation, strategy, analytics, competitor/market research, and audience engagement) over a FastAPI backend.
- Fine-tuned an 8B-parameter LLM (Qwen) via knowledge distillation and QLoRA (4-bit quantized LoRA) on NVIDIA A100 GPUs (SLURM HPC cluster), then self-hosted it for inference through vLLM and Modal serverless GPUs behind an OpenAI-compatible API.
- Engineered a fault-tolerant LLM gateway (LiteLLM) with automatic multi-provider failover and health-checked routing across self-hosted (Modal, vLLM) and hosted (OpenAI, Anthropic, OpenRouter) backends, with per-agent model policies.
- Built a Retrieval-Augmented Generation (RAG) layer on Supabase pgvector with embedding-based semantic retrieval, grounding agents in brand context, competitor data, and live market signals ingested via Firecrawl and Apify behind our own scraping tool.
- Developed asynchronous task orchestration with Redis + Celery, schema-validated LLM outputs (Pydantic), Dockerized deployment on Google Cloud (GCloud), and Sentry observability; built the frontend in Next.js / React / TypeScript.

Education

B.Sc. Computer Science and Artificial Intelligence

Sep 2022 – June 2026

Zewail City of Science and Technology

Cairo, Egypt

- **Coursework:** Deep Learning, NLP, Computer Vision, Data Mining, Algorithm Design, Explainable AI, Reinforcement Learning, Speech Recognition, MLOps, Big Data, GANs, AI Safety
- **Thesis:** Multi-Agent Orchestration for Social Media Content Automation

Technical Skills

Programming: Python, SQL, C++ (*Familiar with: Java, JavaScript, C#, .NET*)

Machine Learning: PyTorch, Scikit-learn, Hugging Face Transformers

LLMs & Agents: LangChain, LlamaIndex, OpenAI API, Anthropic Claude, CrewAI, DSPy, n8n

RAG & Vector Databases: FAISS, ChromaDB, pgvector, Pinecone, Neo4j, Hybrid Retrieval

MLOps & Production: FastAPI, Docker, GCloud, Modal, vLLM, LiteLLM, Redis, Celery, Vast.ai, PostgreSQL, Git, Ragas

Data Engineering & Big Data: Apache Spark (PySpark), Pandas, distributed ETL, Parquet

Explainability: SHAP, LIME, PDP, Feature Importance Analysis, ICE

Familiar with: TensorFlow, AWS, Kubernetes, MongoDB

Certifications

Machine Learning Specialization – Stanford Online (Andrew Ng) | **Building RAG Applications** – DeepLearning.AI

Leadership

Head of Multimedia Committee | IEEE Zewail City

Sep 2024 – May 2025

- Led team of 8 designers for 30+ technical events; increased engagement by 50%.

References

Dr. Ahmed Sayed Abdelsamea Mohamed – Associate Professor of Applied Mathematics, Director of Academic Advising, Zewail City | *Senior IEEE Member* | aabdelsamea@zewailcity.edu.eg | [Recommendation Letter](#)